Lab Documentation: Programs 23 & 24

Program 23: Draw a Line Using Axis Values from a Text File

AIM

To write a Python program to draw a line using given axis values taken from a text file, with suitable labels and a title.

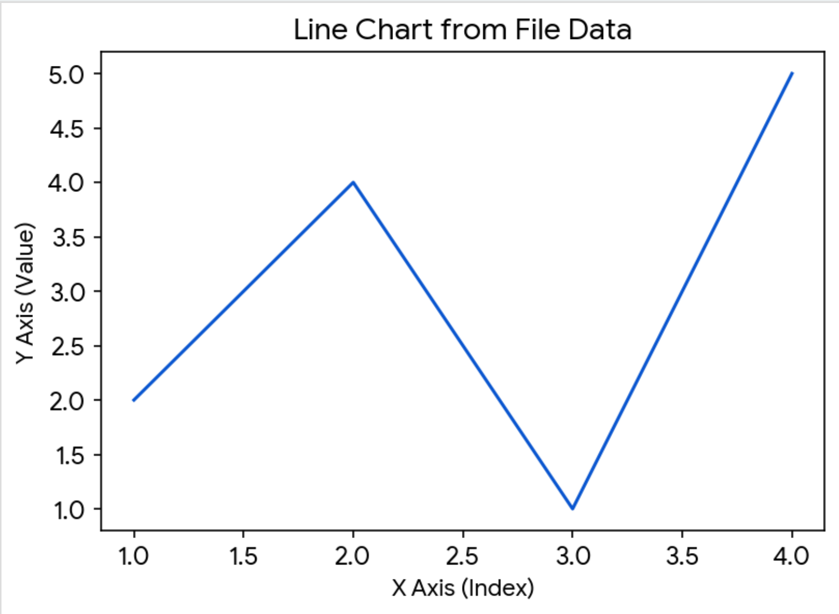
CODE

```
import matplotlib.pyplot as plt

with open('data.txt', 'r') as f:
    x, y = [], []
    for line in f:
        val = line.split()
        x.append(float(val[0]))
        y.append(float(val[1]))

plt.plot(x, y)
plt.xlabel('X Axis (Index)')
plt.ylabel('Y Axis (Value)')
plt.title('Line Chart from File Data')
plt.show()
```

OUTPUT



Result: The coordinates were successfully extracted from the text file and visualized as a line chart.

Program 24: Draw Line Charts of Financial Data

AIM

To write a Python program to draw line charts representing financial data.

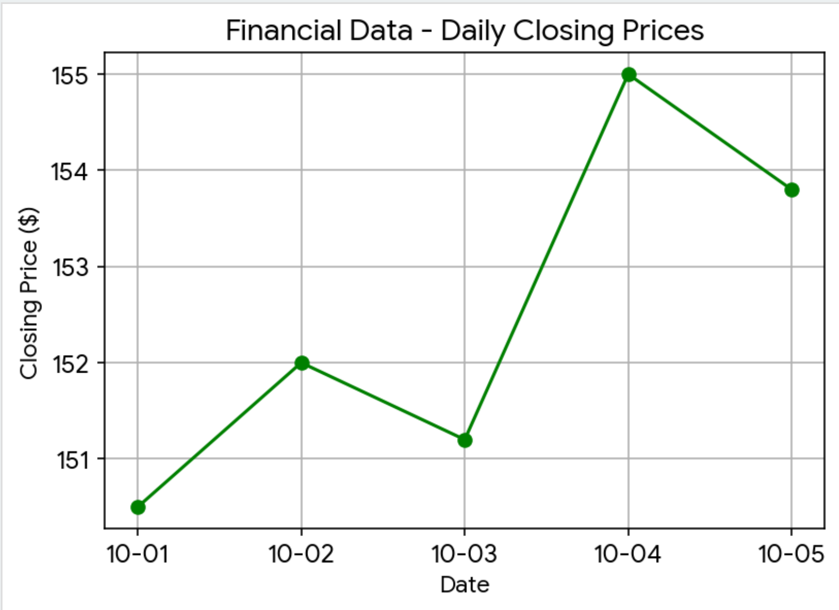
CODE

```
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    'Date': ['10-01', '10-02', '10-03', '10-04', '10-05'],
    'Close_Price': [150.5, 152.0, 151.2, 155.0, 153.8]
})

plt.plot(df['Date'], df['Close_Price'], marker='o', color='green')
plt.xlabel('Date')
plt.ylabel('Closing Price ($)')
plt.title('Financial Data - Daily Closing Prices')
plt.grid(True)
plt.show()
```

OUTPUT



Result: The financial data was successfully mapped to a chronological line chart.